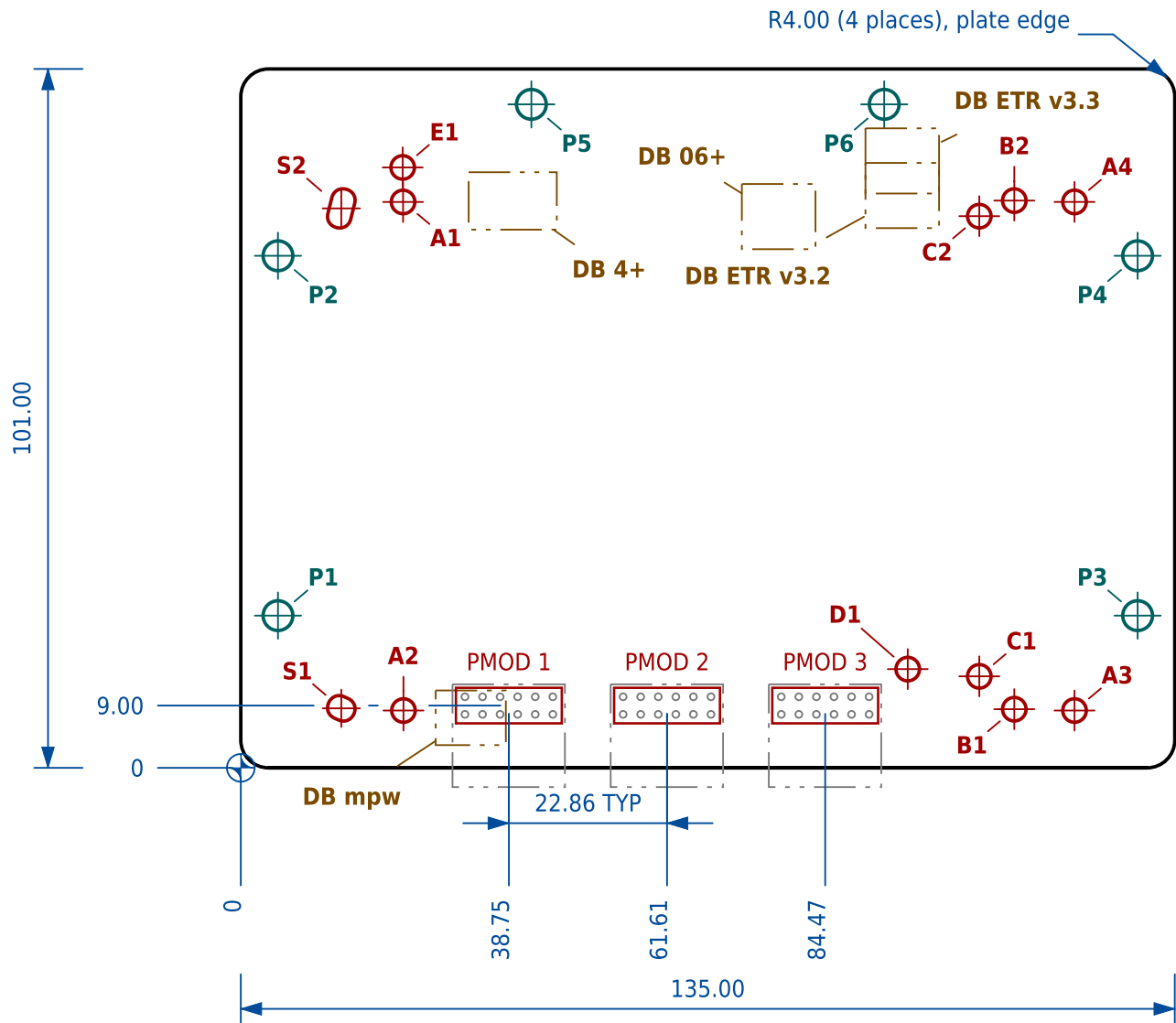




## NOTES

- Viewed from the side the board mounts on.
- A slot row gives its two end centres; DIA/WIDTH is the slot width and LENGTH its overall length.
- USED BY names the board a feature serves and that board's own hole ID, MT1 to MT4. Drawing TT-MP-02 shows each board on the plate.
- A hole's ID letter is the board it is there for, A DB mpw, B DB 4+, C DB 06+, D DB ETR v3.2, E DB ETR v3.3, numbered within it. S is a slot, P a plate fixing. A feature two boards share keeps one ID.
- PMOD 1 to 3 are not machined features; they mark where every board's Pmod host pin fields land.
- Board holes and slots are 3.4 mm, clearance for M3 standoffs. Plate fixings are 4.3 mm, clearance for M4. A slot serves two boards whose holes are too close together to drill separately.

## NOTES (continued)

- Pmod connector bodies overhang the front (lower) edge by 2.78 mm; keep it clear so a peripheral can plug in.
- Cut file: tt-generic-mounting-plate.dxf. Sizes are finished sizes.
- Fit the plate to the chassis before the board: a board overhangs the plate fixings. Thinnest web between features is 1.59 mm, between A1 and E1.

## SOURCES

- Derived from: tinytapeout/boards.py - Board outlines, mounting holes and Pmod host positions for every Tiny Tapeout demo board revision, extracted from the upstream KiCad files.
- Pmod host pitch: <https://mith.ro/pmod-spec/> - Digilent mandate .90 in (22.86 mm) between adjacent host ports on a board edge, which is why one plate can serve every revision.

## BOARD MOUNTING HOLES

| ID | X mm          | Y mm          | DIA/WIDTH mm | LENGTH mm | USED BY                          |
|----|---------------|---------------|--------------|-----------|----------------------------------|
| A1 | 23.45         | 81.78         | 3.40         | -         | DB mpw:MT3, DB ETR v3.2:MT1      |
| A2 | 23.51         | 8.29          | 3.40         | -         | DB mpw:MT1                       |
| A3 | 120.51        | 8.29          | 3.40         | -         | DB mpw:MT2                       |
| A4 | 120.51        | 81.78         | 3.40         | -         | DB mpw:MT4                       |
| B1 | 111.81        | 8.47          | 3.40         | -         | DB 4+:MT2                        |
| B2 | 111.81        | 81.97         | 3.40         | -         | DB 4+:MT4                        |
| C1 | 106.75        | 13.22         | 3.40         | -         | DB 06+:MT2                       |
| C2 | 106.75        | 79.72         | 3.40         | -         | DB 06+:MT4                       |
| D1 | 96.40         | 14.27         | 3.40         | -         | DB ETR v3.2:MT2, DB ETR v3.3:MT2 |
| E1 | 23.40         | 86.77         | 3.40         | -         | DB ETR v3.3:MT1                  |
| S1 | 14.24 / 14.80 | 8.72 / 8.47   | 3.40         | 4.01      | DB 4+:MT1, DB 06+:MT1            |
| S2 | 14.24 / 14.80 | 79.72 / 81.97 | 3.40         | 5.72      | DB 4+:MT3, DB 06+:MT3            |

## PLATE FIXING HOLES

| ID | X mm   | Y mm  | DIA mm |
|----|--------|-------|--------|
| P1 | 5.37   | 22.00 | 4.30   |
| P2 | 5.37   | 74.00 | 4.30   |
| P3 | 129.63 | 22.00 | 4.30   |
| P4 | 129.63 | 74.00 | 4.30   |
| P5 | 42.00  | 95.89 | 4.30   |
| P6 | 93.00  | 95.89 | 4.30   |

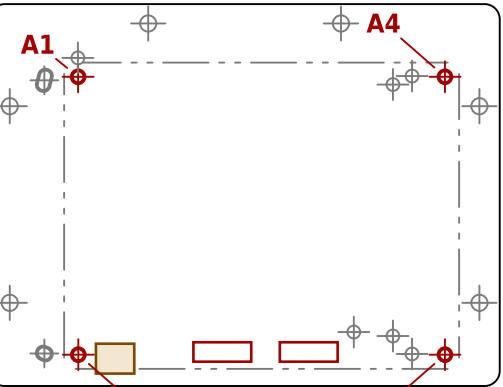
## LEGEND

- Plate edge
- Hole or slot for a demo board fastener
- Plate fixing, into the chassis
- Slot axis
- Pmod host pin field, where a board's hosts land
- Pmod connector body, overhanging the front edge
- USB-C connector, one position per board version
- Dimension, extension and leader

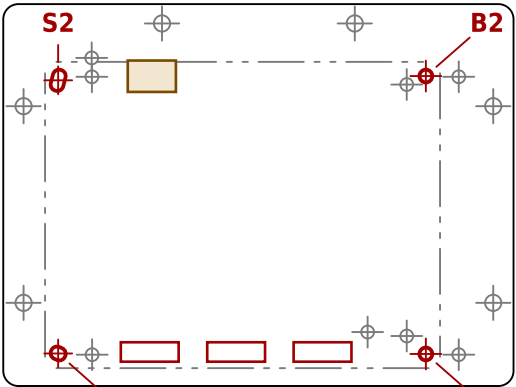
## TT GENERIC MOUNTING PLATE

Accepts every demo board revision, Pmod hosts fixed in place

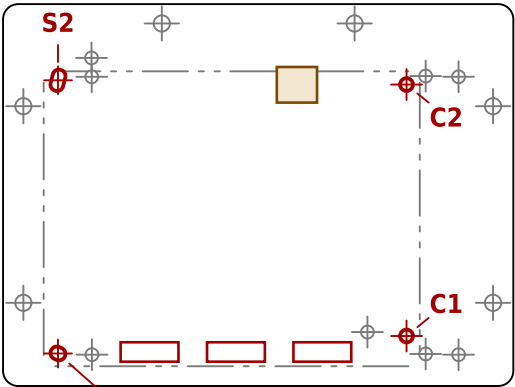
|                            |                  |                                                     |       |
|----------------------------|------------------|-----------------------------------------------------|-------|
| DRAWN                      | VERSION          | UNITS                                               | SCALE |
| generated                  | v0.0-114-g36eaa9 | mm                                                  | 1:1   |
| SIZE                       | SHEET            | DRAWING NO                                          | REV   |
| A3                         | 1 OF 2           | TT-MP-01                                            | A     |
| MATERIAL                   |                  | GENERAL TOLERANCE                                   |       |
| 3 mm acrylic or 1.6 mm FR4 |                  | edge +/-0.20 hole pos +/-0.10 allow for cutter kerf |       |



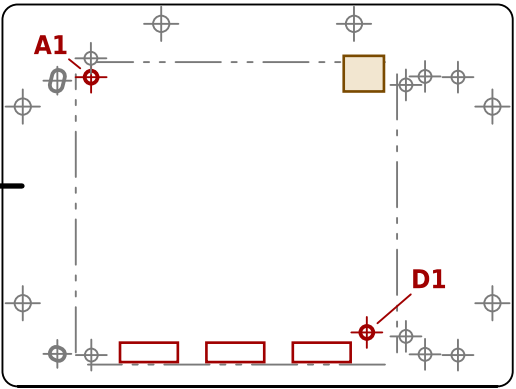
**DB mpw**  
TT02, TT03  
board rev tt123-v2.2.5, tt123-v2.2.6



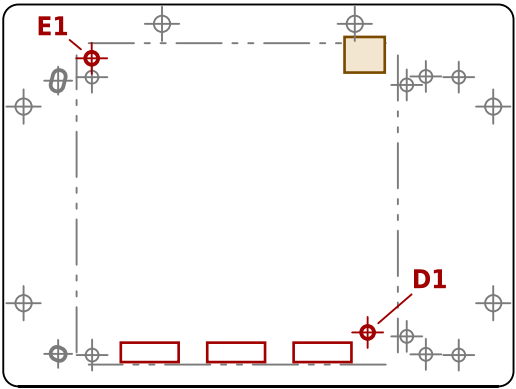
**DB 4+**  
TT03p5, TT04, TT05  
board rev v1.2.1, v1.2.2, v1.2.3



**DB 06+**  
TT06, TT07, TT08  
board rev v2.0.1, v2.1.0, v2.1.2



**DB ETR v3.2**  
TT09, TTSKY25a, TTSKY25b, TTGF0p2  
board rev v3.2



**DB ETR v3.3**  
no shipped shuttle yet  
board rev v3.3

### NOTES

- Hole IDs are those of drawing TT-MP-01, which governs every dimension.
- Revisions in a group share mounting holes and Pmod host positions exactly; they may differ elsewhere. v2.1.2's USB-C, for instance, is 0.9 mm from v2.0.1's.
- dX and dY place the board: add them to a coordinate in the board's own frame to get a plate coordinate.

### BOARD PLACEMENT ON THE PLATE

| BOARD       | SHUTTLES                          | dX    | dY   | PMODS   | HOLES USED     |
|-------------|-----------------------------------|-------|------|---------|----------------|
| DB mpw      | TT02, TT03                        | 19.76 | 4.54 | 2, 3    | A1, A2, A3, A4 |
| DB 4+       | TT03p5, TT04, TT05                | 11.05 | 4.72 | 1, 2, 3 | B1, B2, S1, S2 |
| DB 06+      | TT06, TT07, TT08                  | 10.74 | 5.22 | 1, 2, 3 | C1, C2, S1, S2 |
| DB ETR v3.2 | TT09, TTSKY25a, TTSKY25b, TTGF0p2 | 19.40 | 5.77 | 1, 2, 3 | A1, D1         |
| DB ETR v3.3 | not yet shipped                   | 19.40 | 5.77 | 1, 2, 3 | D1, E1         |

### USB-C CONNECTOR POSITION ON THE PLATE

| BOARD       | X EXTENT mm     | Y EXTENT mm    |
|-------------|-----------------|----------------|
| DB mpw      | 28.17 to 38.31  | 3.29 to 11.19  |
| DB 4+       | 32.94 to 45.67  | 77.78 to 86.08 |
| DB 06+      | 72.42 to 83.06  | 74.95 to 84.38 |
| DB ETR v3.2 | 90.29 to 100.93 | 78.00 to 87.42 |
| DB ETR v3.3 | 90.29 to 100.93 | 83.00 to 92.42 |

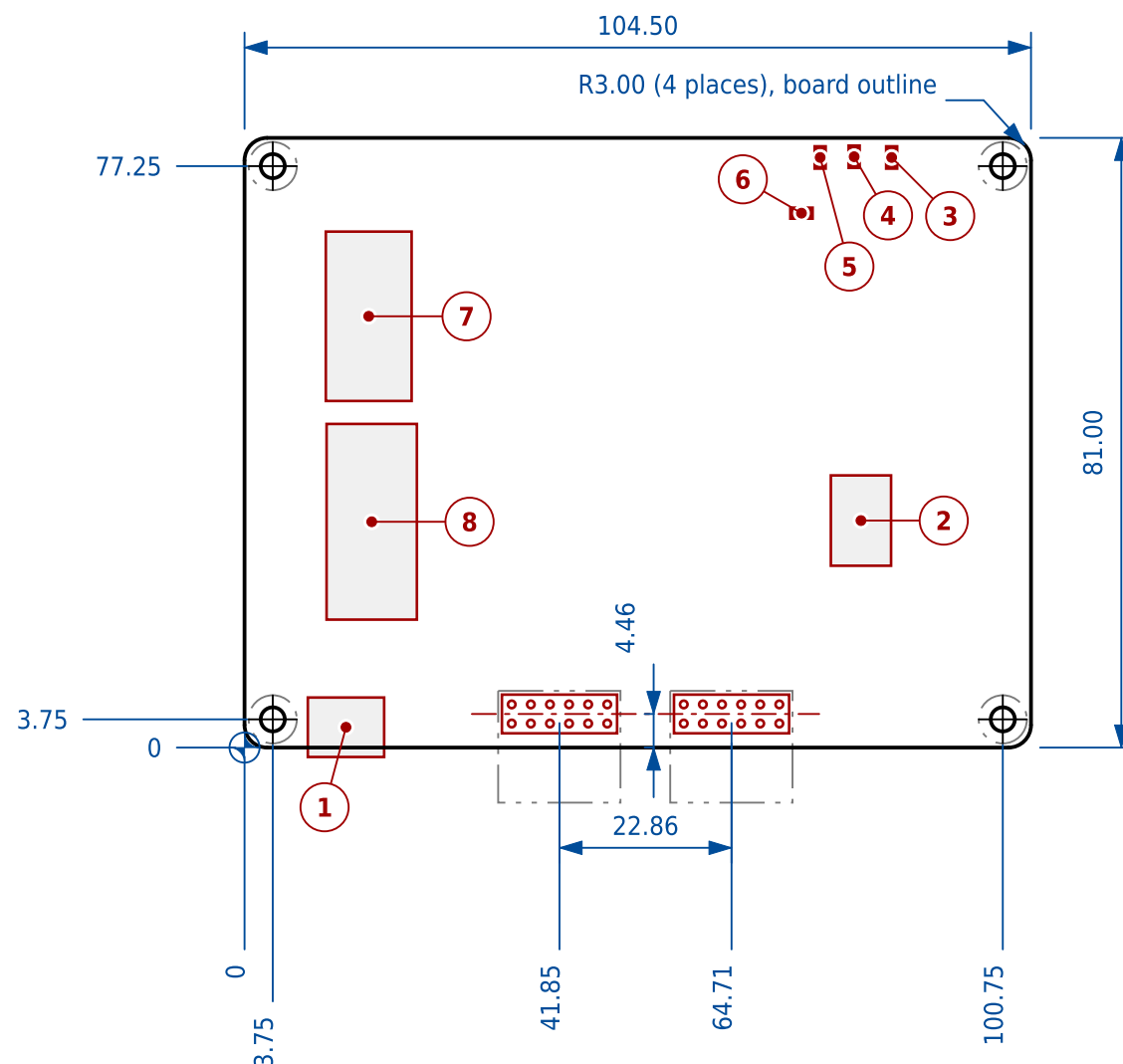
### LEGEND

- Plate edge
- Hole or slot this revision uses
- Holes it does not use, and its board outline
- USB-C connector on this revision

### TT MOUNTING PLATE FITTING GUIDE

Which holes each demo board revision uses

|                   |                   |                                                   |       |
|-------------------|-------------------|---------------------------------------------------|-------|
| DRAWN             | VERSION           | UNITS                                             | SCALE |
| generated         | v0.0-114-g36eaad9 | mm                                                | 1:2   |
| SIZE              | SHEET             | DRAWING NO                                        | REV   |
| A3                | 2 OF 2            | TT-MP-02                                          | A     |
| MATERIAL          |                   | GENERAL TOLERANCE                                 |       |
| - not a made part |                   | reference only - TT-MP-01 governs every dimension |       |



## NOTES

1. Viewed from the component side.
2. Ships in TT02 Dev Kit and TT03 Dev Kit.
3. Hole IDs are the KiCad reference designators.
4. Named "TinyTapeout 1,2,3 Demo Board" in KiCad, but no PCB served TT01, which was a bare-die trial run.
5. The revisions covered are mechanically identical; only the electrical design differs.
6. Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-01, registers against it.
7. Assembled envelope, connector overhang included: 104.50 x 82.25 mm.

## SOURCES

1. KiCad board file:  
[https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad\\_pcb](https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad_pcb) @ 303509a - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.5, dated 2023-11-13
2. KiCad board file:  
[https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad\\_pcb](https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad_pcb) @ 3d721a6 - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.6, dated 2024-02-14
3. Shuttle mapping: <https://mith.ro/tt-demoboards/> - used by: TT02, TT03

## HOLE SCHEDULE

| ID  | X mm   | Y mm  | DIA mm | KEEPOUT mm |
|-----|--------|-------|--------|------------|
| MT1 | 3.75   | 3.75  | 3.20   | 6.40       |
| MT2 | 100.75 | 3.75  | 3.20   | 6.40       |
| MT3 | 3.75   | 77.25 | 3.20   | 6.40       |
| MT4 | 100.75 | 77.25 | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| PMOD A | J3  | bottom | 41.850 | 4.465 | 48.200    | 5.735     |
| PMOD B | J9  | bottom | 64.710 | 4.465 | 71.060    | 5.735     |

## FEATURE SCHEDULE

| #  | FEATURE                                        | X EXTENT mm    | Y EXTENT mm    |
|----|------------------------------------------------|----------------|----------------|
| 1  | USB-C power / control                          | 8.41 to 18.55  | -1.25 to 6.65  |
| 2  | 7-segment display                              | 77.91 to 85.91 | 24.16 to 36.16 |
| 3  | LED D1                                         | 85.25 to 86.72 | 76.92 to 79.88 |
| 4  | LED D2                                         | 80.27 to 81.73 | 77.02 to 79.98 |
| 5  | LED D3                                         | 75.75 to 77.22 | 76.92 to 79.88 |
| 6  | LED D4                                         | 72.52 to 75.48 | 70.27 to 71.73 |
| 7  | 8-way input DIP switch                         | 10.80 to 22.20 | 46.05 to 68.55 |
| 8  | 9-way DIP switch SW2                           | 10.90 to 22.90 | 17.00 to 43.00 |
| 9  | Side Pmod, first position - not on this board  | -              | -              |
| 10 | Side Pmod, second position - not on this board | -              | -              |
| 11 | Side Pmod, third position - not on this board  | -              | -              |

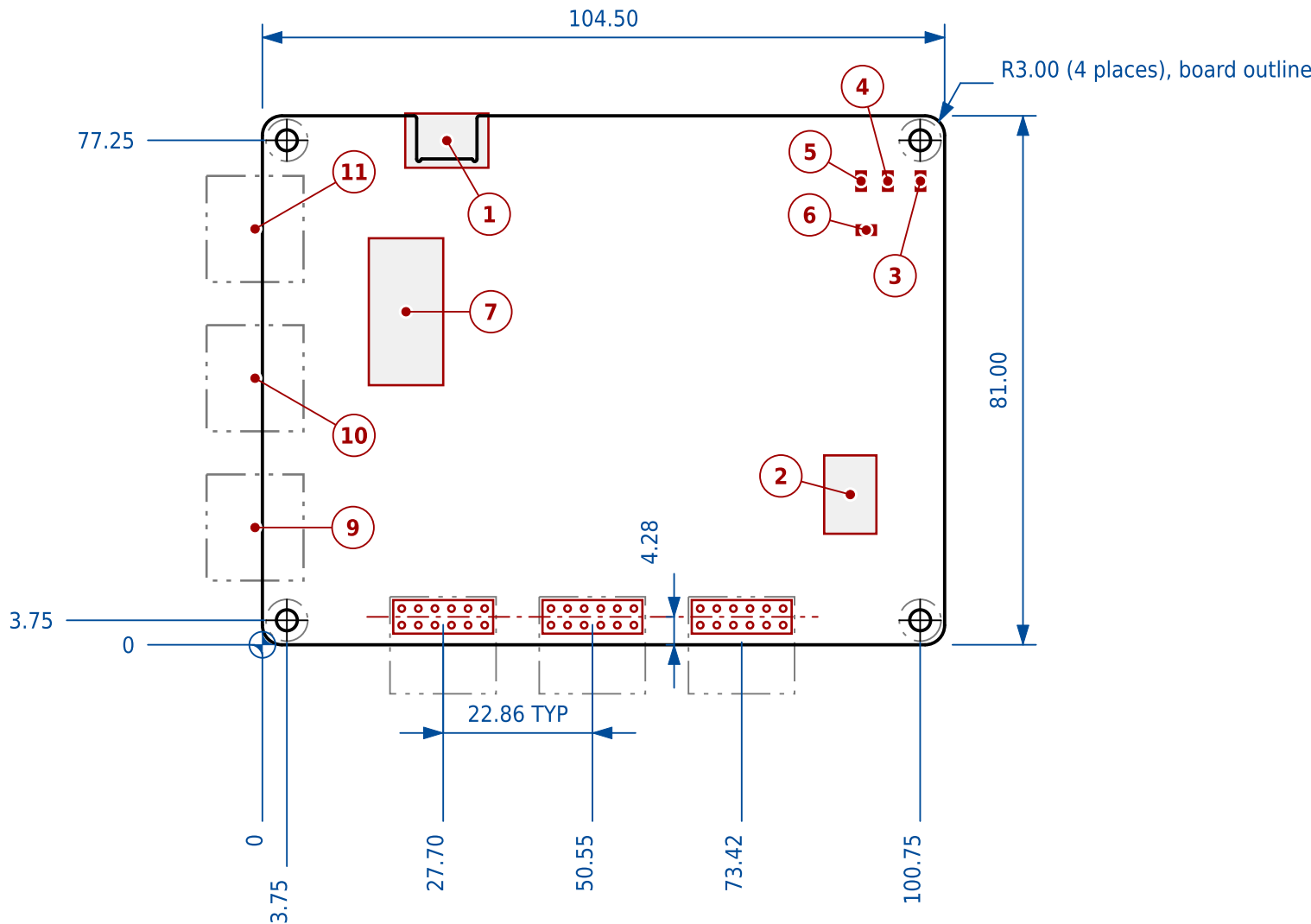
## LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB MPW V2.2.5 / DB MPW V2.2.6

mpw-mb1 rev 2.2.5 and 2.2.6

|                  |                  |                                                |       |
|------------------|------------------|------------------------------------------------|-------|
| DRAWN            | VERSION          | UNITS                                          | SCALE |
| generated        | v0.0-114-g36eaa9 | mm                                             | 1:1   |
| SIZE             | SHEET            | DRAWING NO                                     | REV   |
| A3               | 1 OF 1           | TT-DB-01                                       | A     |
| MATERIAL         |                  | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                  | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |



## NOTES

1. Viewed from the component side.
2. Ships in TT03p5 ASIC IC kit, TT04 Dev Kit and TT05 Dev Kit.
3. Hole IDs are the KiCad reference designators.
4. TT03p5's board is recorded as rev 1.2.1 but unconfirmed: the TT03p5 render is labelled v1.1.2.
5. v1.2.2c is Tiny Tapeout's name for rev 1.2.3, an assembly variant of 1.2.2.
6. The revisions covered are mechanically identical; only the electrical design differs.
7. Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-01, registers against it.
8. Assembled envelope, connector overhang included: 104.50 x 81.35 mm.

## NOTES (continued)

9. The notch in the upper edge is a recess for the USB-C shell.

## SOURCES

1. KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ 8aad3f8 - title block: TinyTapeout 4+ Demoboard rev 1.2.1, dated 2024-04-12
2. KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ cfdd80d7b - title block: TinyTapeout 4+ Demoboard rev 1.2.2, dated 2024-04-26
3. KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ a88cbc08b - title block: TinyTapeout 4+ Demoboard rev 1.2.3, dated 2024-06-11
4. Shuttle mapping: <https://mith.ro/tt-demoboards/> - used by: TT03p5, TT04, TT05

## HOLE SCHEDULE

| ID  | X mm   | Y mm  | DIA mm | KEEPOUT mm |
|-----|--------|-------|--------|------------|
| MT1 | 3.75   | 3.75  | 3.20   | 6.40       |
| MT2 | 100.75 | 3.75  | 3.20   | 6.40       |
| MT3 | 3.75   | 77.25 | 3.20   | 6.40       |
| MT4 | 100.75 | 77.25 | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| INPUT  | J3  | bottom | 27.695 | 4.275 | 34.045    | 5.545     |
| BIDIR  | J5  | bottom | 50.555 | 4.275 | 56.905    | 5.545     |
| OUTPUT | J6  | bottom | 73.415 | 4.275 | 79.765    | 5.545     |

## FEATURE SCHEDULE

| #  | FEATURE                               | X EXTENT mm      | Y EXTENT mm    |
|----|---------------------------------------|------------------|----------------|
| 1  | USB-C power / control                 | 21.88 to 34.62   | 73.05 to 81.35 |
| 2  | 7-segment display                     | 86.05 to 94.05   | 17.00 to 29.00 |
| 3  | LED D1                                | 100.07 to 101.53 | 69.52 to 72.48 |
| 4  | LED D2                                | 95.07 to 96.53   | 69.52 to 72.48 |
| 5  | LED D3                                | 90.97 to 92.43   | 69.52 to 72.48 |
| 6  | LED D4                                | 91.02 to 93.98   | 62.77 to 64.23 |
| 7  | 8-way input DIP switch                | 16.30 to 27.70   | 39.75 to 62.25 |
| 8  | Second DIP switch - not on this board | -                | -              |
| 9  | Side Pmod J12 (not fitted)            | -8.55 to 6.30    | 9.83 to 26.08  |
| 10 | Side Pmod J13 (not fitted)            | -8.55 to 6.30    | 32.69 to 48.94 |
| 11 | Side Pmod J14 (not fitted)            | -8.55 to 6.30    | 55.55 to 71.80 |

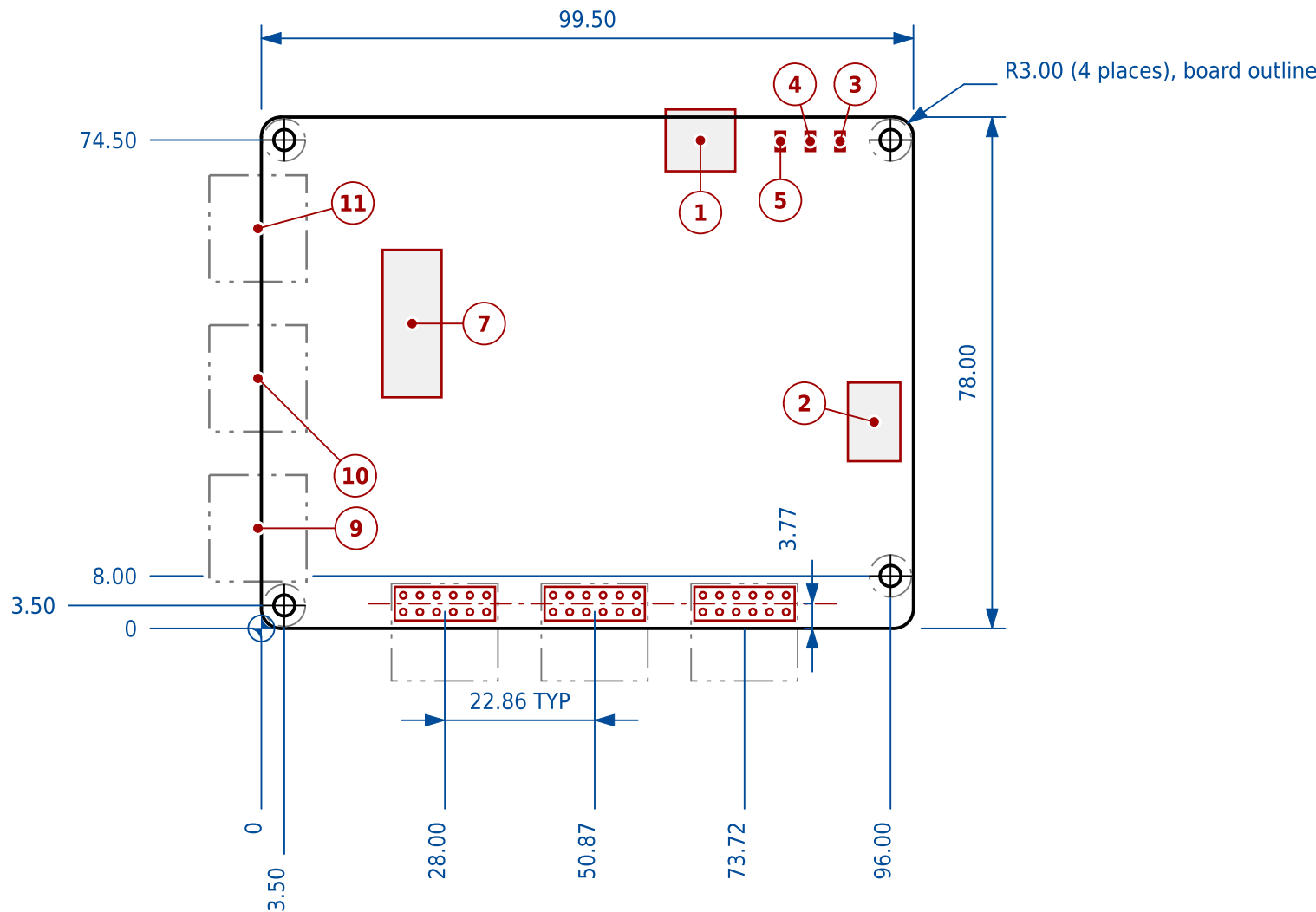
## LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB 4+ V1.2.1 / DB 4+ V1.2.2 / DB 4+ V1.2.2C

tinytapeout-demo rev 1.2.1, 1.2.2 and 1.2.3

|                  |                   |                                                |       |
|------------------|-------------------|------------------------------------------------|-------|
| DRAWN            | VERSION           | UNITS                                          | SCALE |
| generated        | v0.0-114-g36eaad9 | mm                                             | 1:1   |
| SIZE             | SHEET             | DRAWING NO                                     | REV   |
| A3               | 1 OF 1            | TT-DB-02                                       | A     |
| MATERIAL         |                   | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                   | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |



## NOTES

1. Viewed from the component side.
2. Ships in TT06 Dev Kit and TT07 Dev Kit.
3. Hole IDs are the KiCad reference designators.
4. The revisions covered are mechanically identical; only the electrical design differs.
5. Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-01, registers against it.
6. Assembled envelope, connector overhang included: 99.50 x 79.15 mm.

## SOURCES

1. KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ 292760e1f - title block: TinyTapeout 06+ Demoboard rev 2.0.1, dated 2024-09-05
2. KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ a799acb38 - title block: TinyTapeout 06+ Demoboard rev 2.1.0, dated 2024-12-10
3. Shuttle mapping: <https://mith.ro/tt-demoboards/> - used by: TT06, TT07

## HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm | KEEPOUT mm |
|-----|-------|-------|--------|------------|
| MT1 | 3.50  | 3.50  | 3.20   | 6.40       |
| MT2 | 96.00 | 8.00  | 3.20   | 6.40       |
| MT3 | 3.50  | 74.50 | 3.20   | 6.40       |
| MT4 | 96.00 | 74.50 | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| INPUT  | J3  | bottom | 28.005 | 3.775 | 34.355    | 5.045     |
| BIDIR  | J5  | bottom | 50.865 | 3.775 | 57.215    | 5.045     |
| OUTPUT | J6  | bottom | 73.725 | 3.775 | 80.075    | 5.045     |

## FEATURE SCHEDULE

| #  | FEATURE                                  | X EXTENT mm    | Y EXTENT mm    |
|----|------------------------------------------|----------------|----------------|
| 1  | USB-C power / control                    | 61.68 to 72.32 | 69.73 to 79.15 |
| 2  | 7-segment display                        | 89.50 to 97.50 | 25.50 to 37.50 |
| 3  | LED D1                                   | 87.57 to 89.03 | 72.82 to 75.78 |
| 4  | LED D2                                   | 83.02 to 84.48 | 72.82 to 75.78 |
| 5  | LED D3                                   | 78.47 to 79.93 | 72.82 to 75.78 |
| 6  | Fourth indicator LED - not on this board | -              | -              |
| 7  | 8-way input DIP switch                   | 18.50 to 27.50 | 35.25 to 57.75 |
| 8  | Second DIP switch - not on this board    | -              | -              |
| 9  | Side Pmod J12 (not fitted)               | -7.95 to 6.90  | 7.15 to 23.40  |
| 10 | Side Pmod J13 (not fitted)               | -7.95 to 6.90  | 30.01 to 46.26 |
| 11 | Side Pmod J14 (not fitted)               | -7.95 to 6.90  | 52.87 to 69.12 |

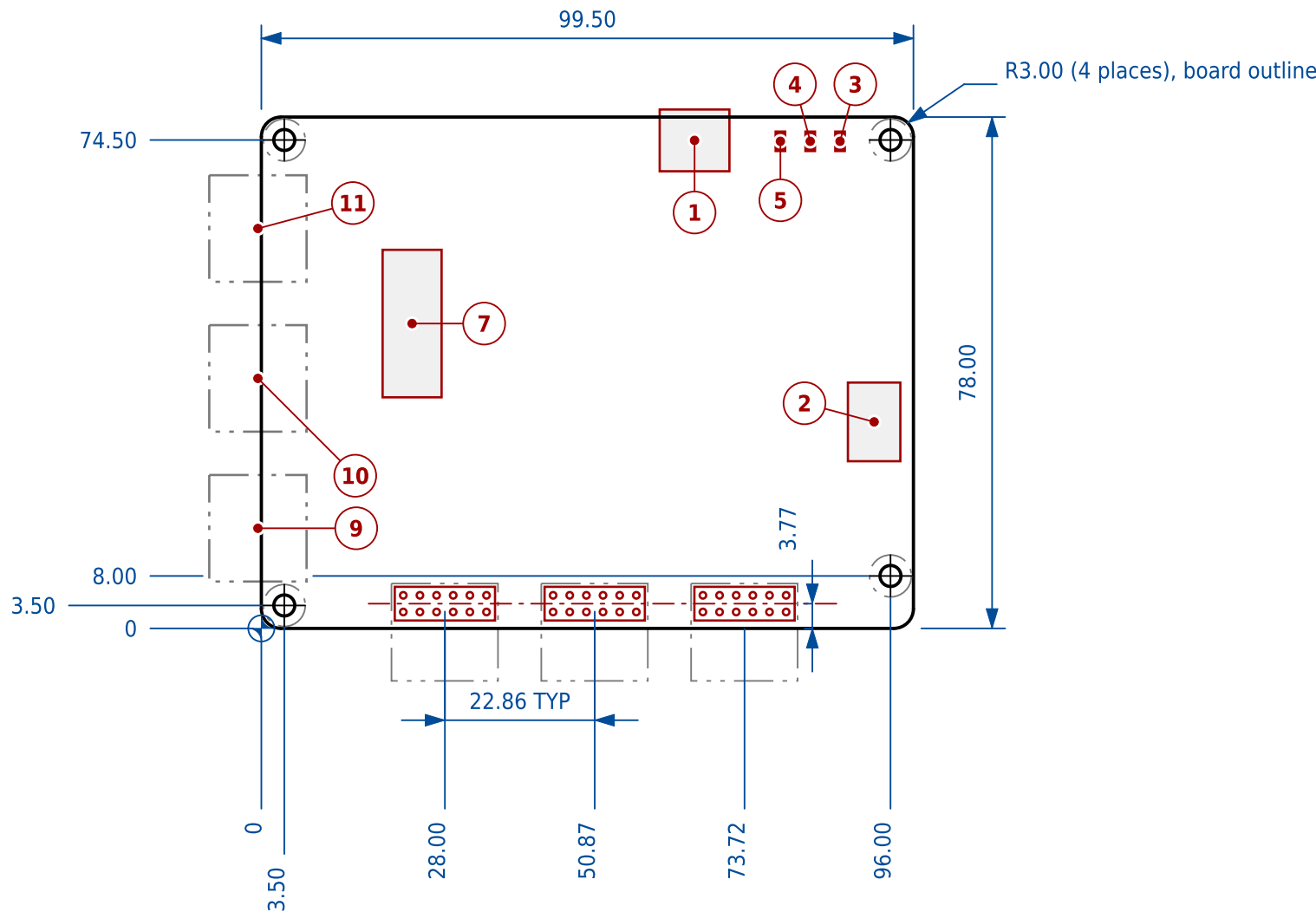
## LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB 06+ V2.0.1 / DB 06+ V2.1.0

tinytapeout-demo rev 2.0.1 and 2.1.0

|                  |                   |                                                |       |
|------------------|-------------------|------------------------------------------------|-------|
| DRAWN            | VERSION           | UNITS                                          | SCALE |
| generated        | v0.0-114-g36eaad9 | mm                                             | 1:1   |
| SIZE             | SHEET             | DRAWING NO                                     | REV   |
| A3               | 1 OF 1            | TT-DB-03                                       | A     |
| MATERIAL         |                   | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                   | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |



## NOTES

- Viewed from the component side.
- Ships in TT08 Dev Kit and TT08 Dev Kit - CoB edition.
- Hole IDs are the KiCad reference designators.
- Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-01, registers against it.
- Assembled envelope, connector overhang included: 99.50 x 79.15 mm.

## SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb>  
tinytapeout-demo.kicad\_pcb @ 028a51b1e - title block:  
TinyTapeout 06+ Demoboard rev 2.1.2, dated 2025-02-11
- Shuttle mapping: <https://mith.ro/tt-demoboards/> - row "DB 06+ v2.1.2", Used by: TT08.

## HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm | KEEPOUT mm |
|-----|-------|-------|--------|------------|
| MT1 | 3.50  | 3.50  | 3.20   | 6.40       |
| MT2 | 96.00 | 8.00  | 3.20   | 6.40       |
| MT3 | 3.50  | 74.50 | 3.20   | 6.40       |
| MT4 | 96.00 | 74.50 | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| INPUT  | J3  | bottom | 28.005 | 3.775 | 34.355    | 5.045     |
| BIDIR  | J5  | bottom | 50.865 | 3.775 | 57.215    | 5.045     |
| OUTPUT | J6  | bottom | 73.725 | 3.775 | 80.075    | 5.045     |

## FEATURE SCHEDULE

| #  | FEATURE                                  | X EXTENT mm    | Y EXTENT mm    |
|----|------------------------------------------|----------------|----------------|
| 1  | USB-C power / control                    | 60.78 to 71.42 | 69.73 to 79.15 |
| 2  | 7-segment display                        | 89.50 to 97.50 | 25.50 to 37.50 |
| 3  | LED D1                                   | 87.57 to 89.03 | 72.82 to 75.78 |
| 4  | LED D2                                   | 83.02 to 84.48 | 72.82 to 75.78 |
| 5  | LED D3                                   | 78.47 to 79.93 | 72.82 to 75.78 |
| 6  | Fourth indicator LED - not on this board | -              | -              |
| 7  | 8-way input DIP switch                   | 18.50 to 27.50 | 35.25 to 57.75 |
| 8  | Second DIP switch - not on this board    | -              | -              |
| 9  | Side Pmod J12 (not fitted)               | -7.95 to 6.90  | 7.15 to 23.40  |
| 10 | Side Pmod J13 (not fitted)               | -7.95 to 6.90  | 30.01 to 46.26 |
| 11 | Side Pmod J14 (not fitted)               | -7.95 to 6.90  | 52.87 to 69.12 |

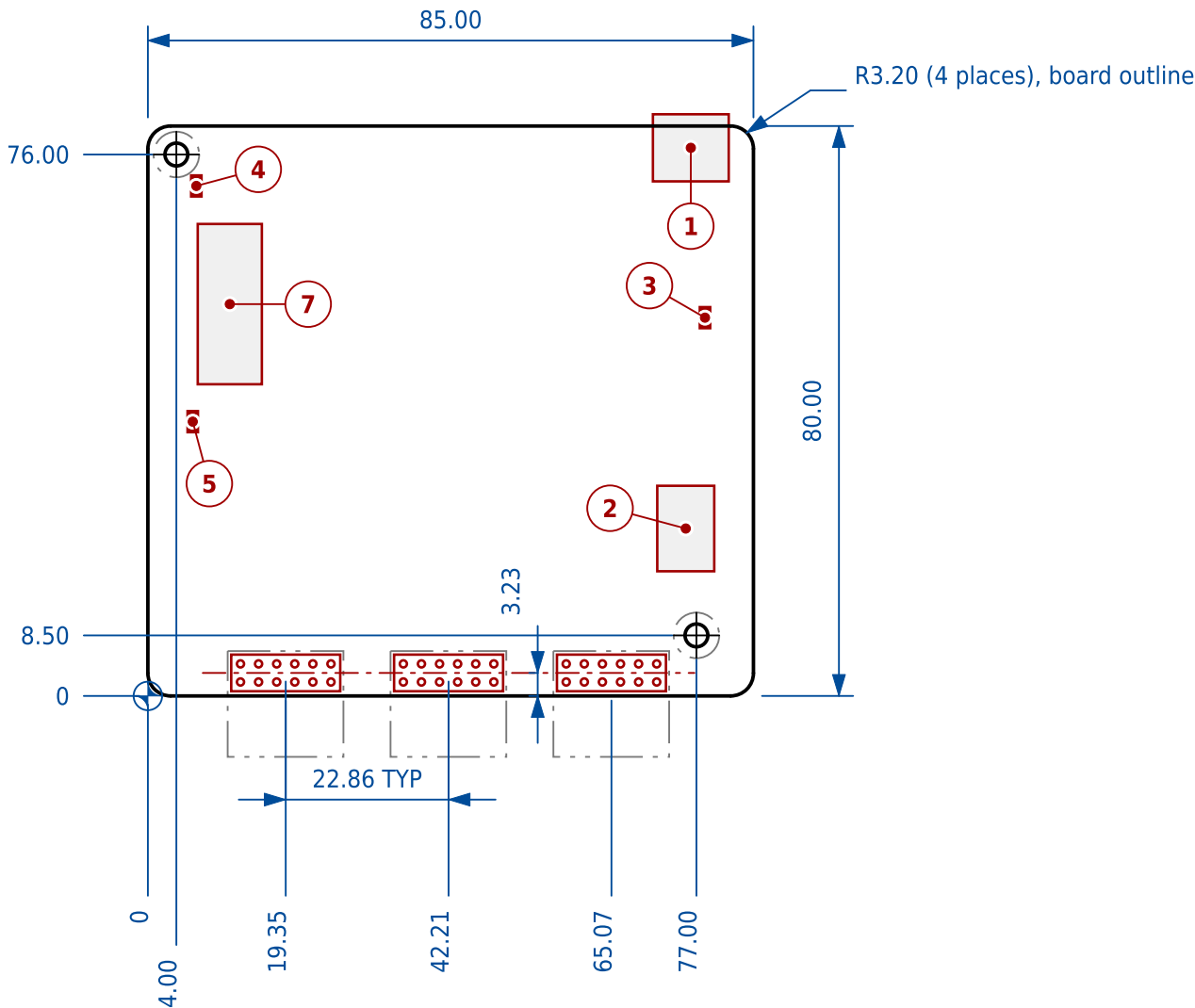
## LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB 06+ V2.1.2

tinytapeout-demo rev 2.1.2

|                  |                  |                                                |       |
|------------------|------------------|------------------------------------------------|-------|
| DRAWN            | VERSION          | UNITS                                          | SCALE |
| generated        | v0.0-114-g36eaa9 | mm                                             | 1:1   |
| SIZE             | SHEET            | DRAWING NO                                     | REV   |
| A3               | 1 OF 1           | TT-DB-04                                       | A     |
| MATERIAL         |                  | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                  | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |



## HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm | KEEPOUT mm |
|-----|-------|-------|--------|------------|
| MT1 | 4.00  | 76.00 | 3.20   | 6.40       |
| MT2 | 77.00 | 8.50  | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| INPUT  | J11 | bottom | 19.350 | 3.230 | 25.700    | 4.500     |
| BIDIR  | J12 | bottom | 42.210 | 3.230 | 48.560    | 4.500     |
| OUTPUT | J13 | bottom | 65.070 | 3.230 | 71.420    | 4.500     |

## FEATURE SCHEDULE

| #  | FEATURE                                        | X EXTENT mm    | Y EXTENT mm    |
|----|------------------------------------------------|----------------|----------------|
| 1  | USB-C power / control                          | 70.89 to 81.53 | 72.23 to 81.65 |
| 2  | 7-segment display                              | 71.50 to 79.50 | 17.50 to 29.50 |
| 3  | LED D3                                         | 77.47 to 78.93 | 51.62 to 54.58 |
| 4  | LED D4                                         | 6.07 to 7.53   | 70.11 to 73.07 |
| 5  | LED D5                                         | 5.57 to 7.03   | 37.02 to 39.98 |
| 6  | Fourth indicator LED - not on this board       | -              | -              |
| 7  | 8-way input DIP switch                         | 7.00 to 16.00  | 43.75 to 66.25 |
| 8  | Second DIP switch - not on this board          | -              | -              |
| 9  | Side Pmod, first position - not on this board  | -              | -              |
| 10 | Side Pmod, second position - not on this board | -              | -              |
| 11 | Side Pmod, third position - not on this board  | -              | -              |

## LEGEND

|  |                                    |
|--|------------------------------------|
|  | Board outline                      |
|  | Component body, scheduled          |
|  | Pmod connector body, hole keep-out |
|  | Dimension, extension and leader    |

## NOTES

- Viewed from the component side.
- Ships in TT09 Dev Kit, TTSKY25a Dev Kit, TTSKY25b Dev Kit, TTGF0p2 Dev Kit - P2 edition and FPGA Dev Kit.
- Hole IDs are the KiCad reference designators.
- "DB ETR" is the board's ID in Tiny Tapeout's board revision spreadsheet; the KiCad title block does not carry it.
- Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-01, registers against it.
- Assembled envelope, connector overhang included: 85.00 x 81.65 mm.

## SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ d830790ca - title block: Tiny Tapeout Demoboard v3 rev 3.2, dated 2025-11-12
- Shuttle mapping: <https://mith.ro/tt-demoboards/> - row "DB ETR v3.2", Used by: TT09, TTSKY25a, TTSKY25b, TTGF0p2.

## DB ETR V3.2

tinytapeout-demo rev 3.2

|                  |                  |                                                |       |
|------------------|------------------|------------------------------------------------|-------|
| DRAWN            | VERSION          | UNITS                                          | SCALE |
| generated        | v0.0-114-g36eaa9 | mm                                             | 1:1   |
| SIZE             | SHEET            | DRAWING NO                                     | REV   |
| A3               | 1 OF 1           | TT-DB-05                                       | A     |
| MATERIAL         |                  | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                  | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |

